

# Region-specific incremental feature menus

Reproduction notes for the *Knowledge and Information Systems* submission

Hoa Dinh Nguyen

Posts and Telecommunications Institute of Technology, Hanoi, Vietnam

hoand@ptit.edu.vn

Python implementation for the manuscript *Region-Specific Incremental Feature Menus for Progressive Feature Revelation*. The method partitions the input space, learns an ordered forward-stepwise feature menu in each region, and predicts at every feature budget with nested prefix least-squares models.

## 1 Install

From the project root:

```
pip install -r requirements.txt
```

Python 3.10 or later is required, together with the packages in `requirements.txt` (NumPy, SciPy, scikit-learn). Matplotlib is needed only for figure scripts.

## 2 Data

Public datasets are downloaded once into `data/`:

```
python examples/download_datasets.py
```

See `data/README.tex`. Experiments then run offline from local `data/*.npz` files. Large archives are not part of the GitHub tree.

## 3 Reproduce the paper artifacts

From the project root:

```
python examples/kais_extra_evidence.py
python examples/kais_stability_budget.py
python examples/generate_paper_figures.py
```

- `examples/kais_extra_evidence.py` writes `submission_kais/extra_evidence.json` (regional menu prefixes, Jaccard overlap, localized Lasso-LARS anytime RMSE).

- `examples/kais_stability_budget.py` writes `submission_kais/stability_and_budget.json` and `fig6_measurement_budget.pdf`.
- `examples/generate_paper_figures.py` writes PDF/PNG figures. By default it currently targets `submission_kbs/figures/`; copy outputs to `submission_kais/figures/` if regenerating the KAIS figures in place.

## 4 Project layout

| Path                          | Contents                                                     |
|-------------------------------|--------------------------------------------------------------|
| <code>src/</code>             | Library: menus, selectors, progressive evaluation            |
| <code>examples/</code>        | Loaders, demos, paper experiments                            |
| <code>submission_kais/</code> | KAIS manuscript and upload files (LaTeX)                     |
| <code>submission_kbs/</code>  | Previous <i>Knowledge-Based Systems</i> package (do not mix) |

## 5 License

Research code accompanying the manuscript. Contact Hoa Dinh Nguyen ([hoand@ptit.edu.vn](mailto:hoand@ptit.edu.vn)) for reuse questions until a public repository URL is attached to the paper.